

HR-MultiAgent-Rec — Layered architecture

① Data Layer

Tianchi · Job-SDF · Synthetic · ESCO → Pydantic schemas



② Semantic Encoding

Qwen3-Embedding-0.6B (MPS) → FAISS-CPU flat / IVF



③ Matching

Qwen3-Reranker-0.6B (CPU yes/no) + Bidirectional scoring



④ Multi-Agent

JobAnalyst · CandidateAnalyst · Coordinator · Explainer (async)



⑤ Evaluation

P@K · R@K · nDCG@K · MRR + Cost / Cache Telemetry